ATOMICA

Title: **Running LaMP**

Document type:

Document ID: **EOI**

Author:

Owner:

Approver(s):

Approval date:

Effective date:

Next periodic review date:

Table of Contents

- # Safety

- The Accretech UF200R contains moving parts. Be careful of motion and pinch points when working with the chuck, head stage and transfer robot.

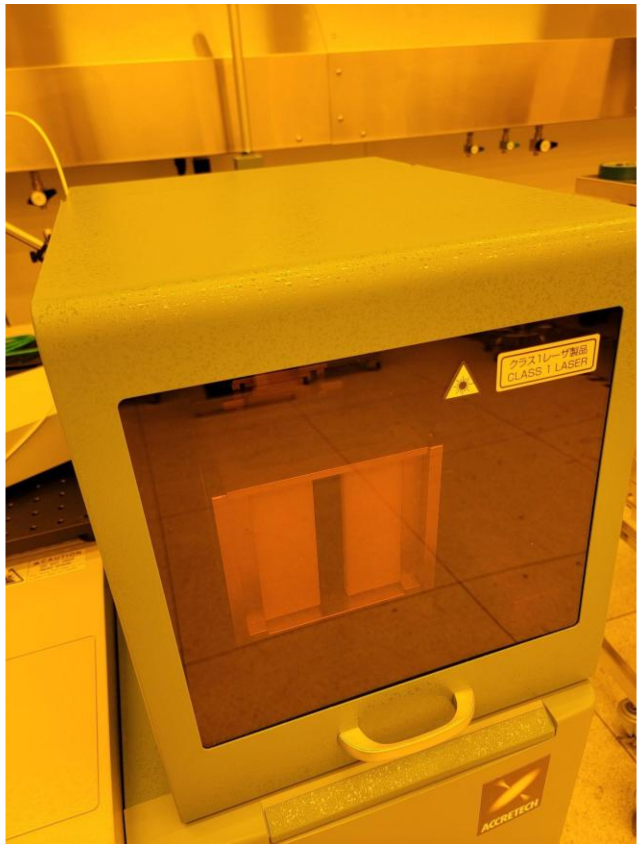
Figure 6a: Mechanical Hazards

- In the event of an emergency, press the red EMO button to stop motion and power off the tool.

Figure 6b: Emergency Off

- The cassette loader uses a Class 1 laser. Never tamper with the cassette loader lock.

Figure 6c: Laser Hazard in Cassette Loader During Operation


クラス1レーザ製品
CLASS 1 LASER
ACCRETECH

Procedure

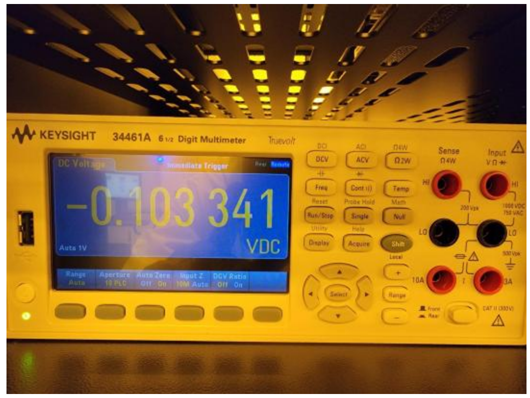
Setup

- Verify Accretech UF200R is powered on without warnings (**Figure 8a**).

Figure 8a: PROBE08 Powered ON

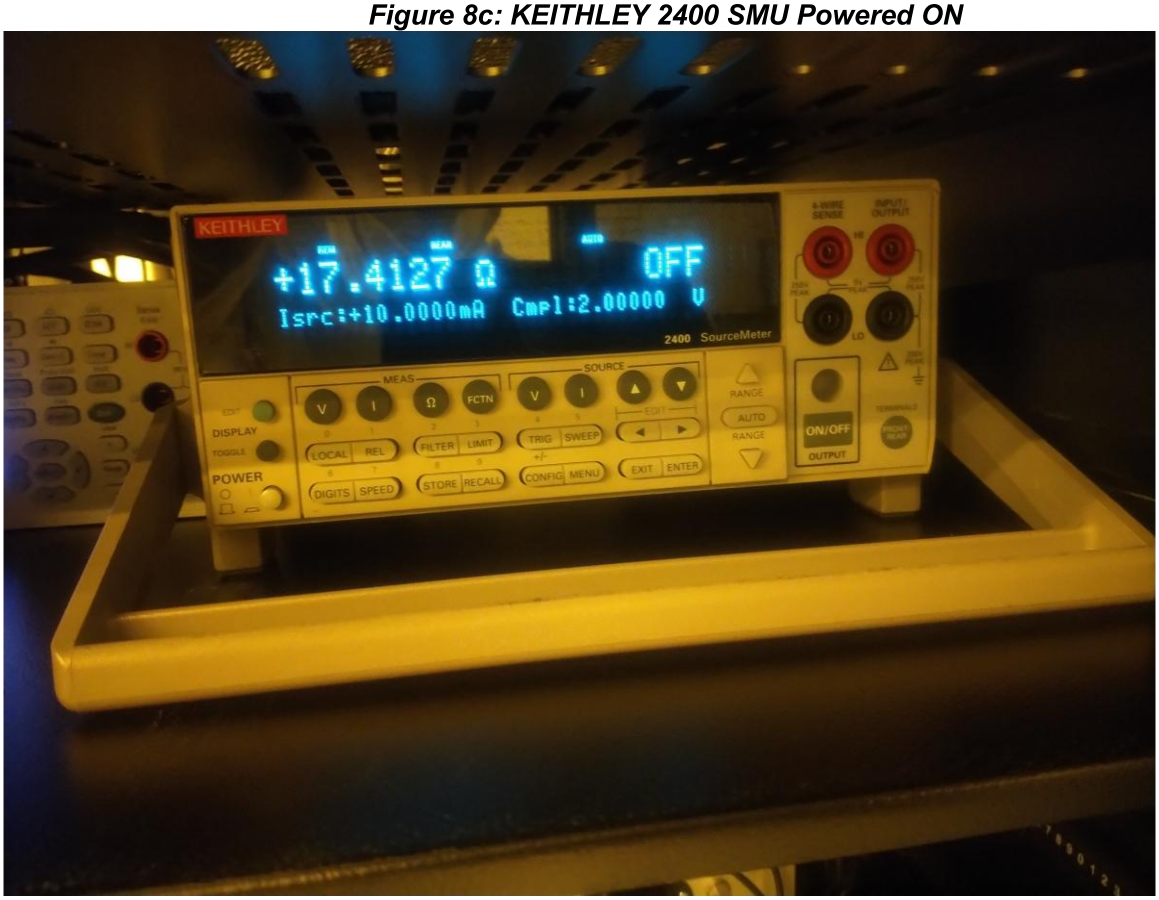
- Verify KEYSIGHT 34461A is powered on (**Figure 8b**).

Figure 8b: KEYSIGHT 34461A DMM Powered ON



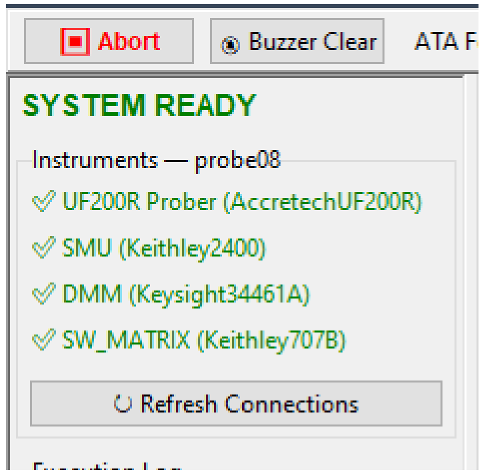
- Verify KEITHLEY 2400 is powered on (**Figure 8c**).

Figure 8c: KEITHLEY 2400 SMU Powered ON



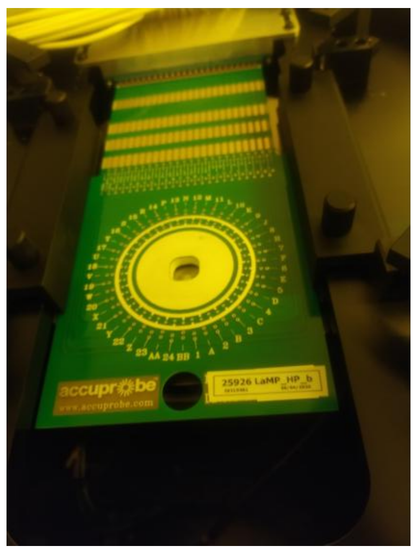
- You can also check all instruments by opening the GUI and verifying the green checkmarks on the left.

Figure 8e: Instrument Connections



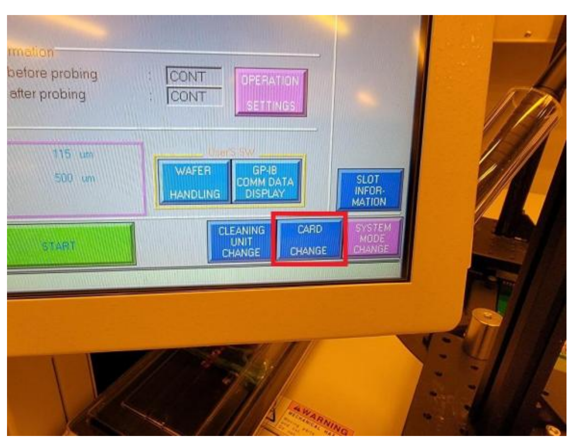
- **Changing Probe Card**

Make sure probe card is LaMP_HP_b. If so, skip this step.



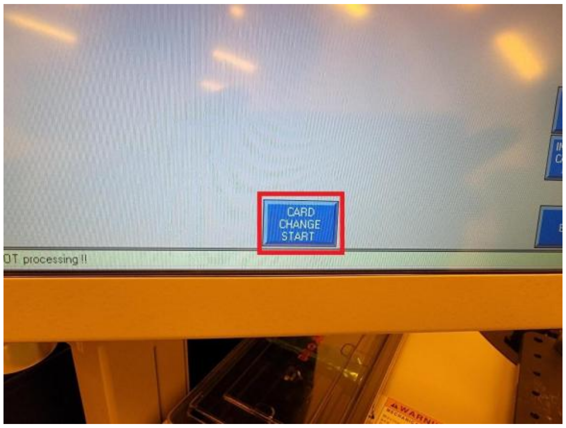
- Within the main production screen, press “CARD CHANGE” (**Figure 8e**).

Figure 8e: CARD CHANGE



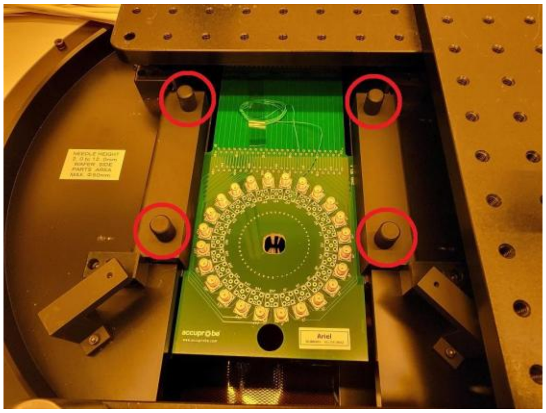
- Press “CARD CHANGE START” (**Figure 8f**).

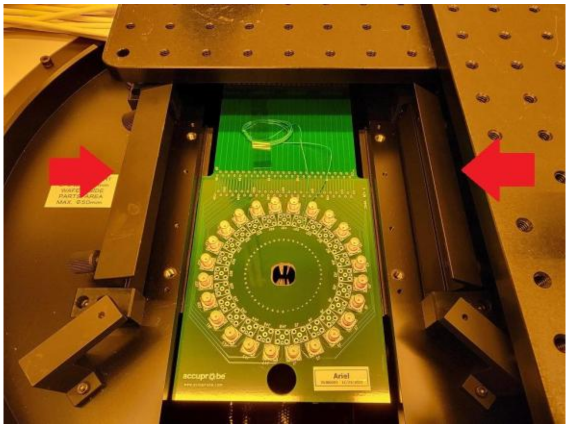
Figure 8f: CARD CHANGE START



- Loosen (4) thumb screws and open clamps (**Figure 8g**).

Figure 8g: Loosen Screws and Open Clamps

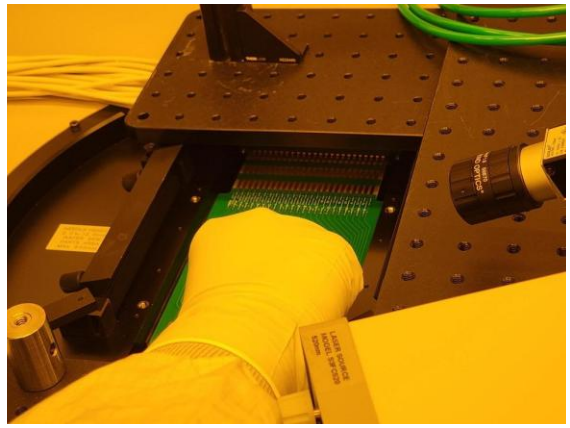




- Use the hole at the front edge of the probe card to carefully pull the probe card out of the connector socket and remove the probe card (**Figure 8h**).

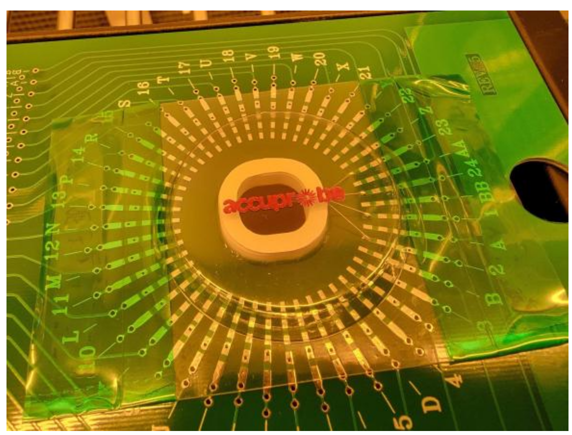
WARNING: Be very careful to avoid damaging the probe needles on the bottom of the probe card.

Figure 8h: Slide Probe Card Out of Edge Connector Socket



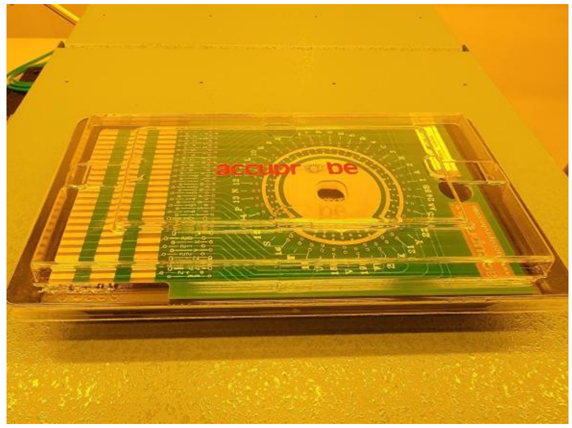
- On the bottom of the probe card, carefully place the plastic probe needle cover over the probe needles and make sure it is secured with tape (**Figure 8i**).

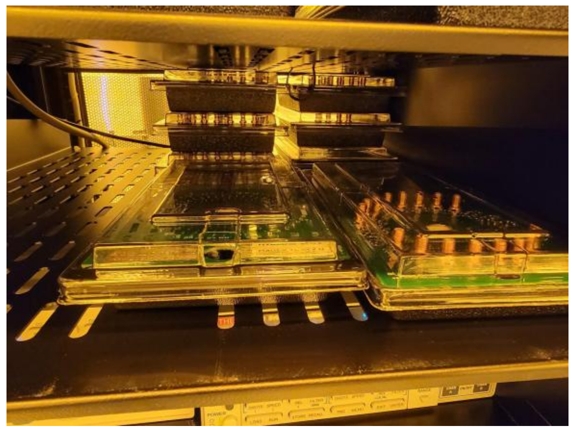
Figure 8i: Probe Needle Cover Centered Over Probe Needles



- Store the old probe card in its case from Accuprobe on the **PROBE08** rack (**Figure 8j**).

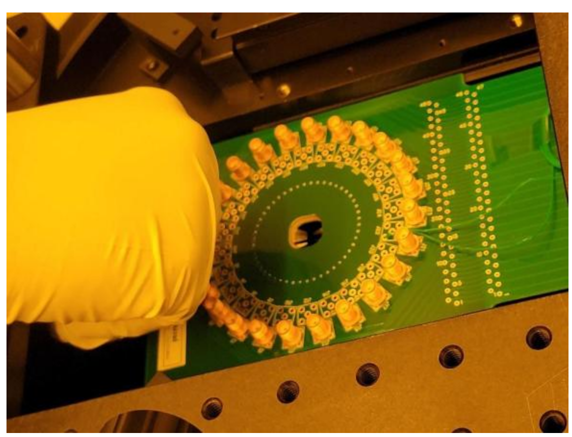
Figure 8j: Store Probe Cards on PROBE08 Rack





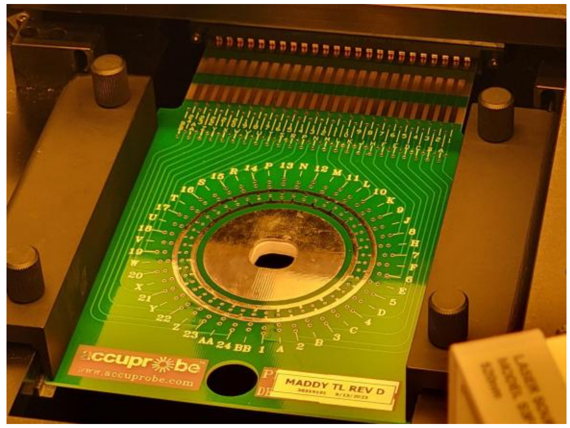
- Carefully remove the plastic probe needle cover from the new probe card and mount it onto the stage rails so that the probe needles are pointing down. Once the new probe card is flat and centered on the rails, use the hole at the front edge of the probe card to push the probe edge connector into the socket (**Figure 8k**).

Figure 8k: Use Rails to Guide Probe Card Into Edge Connector Socket



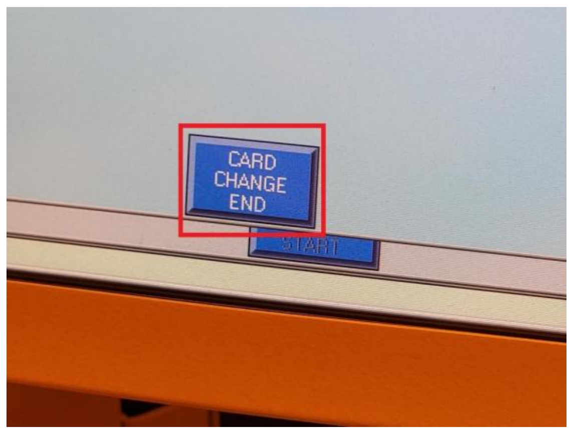
- Bring the clamps down and tighten (4) thumb screws (**Figure 8l**).

Figure 8l: New Probe Card Installed



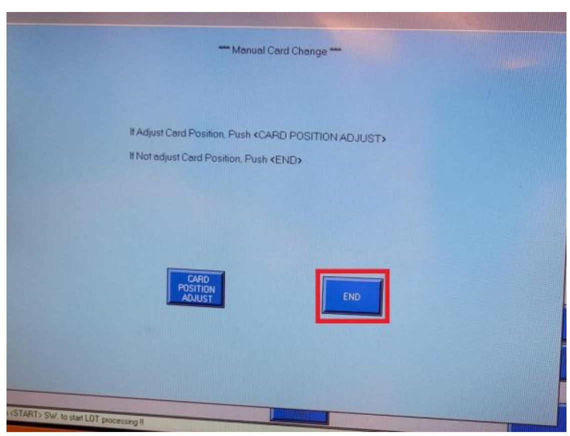
- Press “CARD CHANGE END” (**Figure 8m**).

Figure 8m: CARD CHANGE END



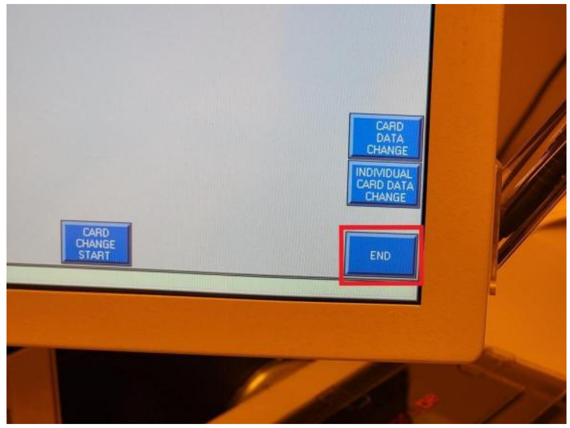
- Press “END” (**Figure 8n**).

Figure 8n: END in Manual Card Change Prompt



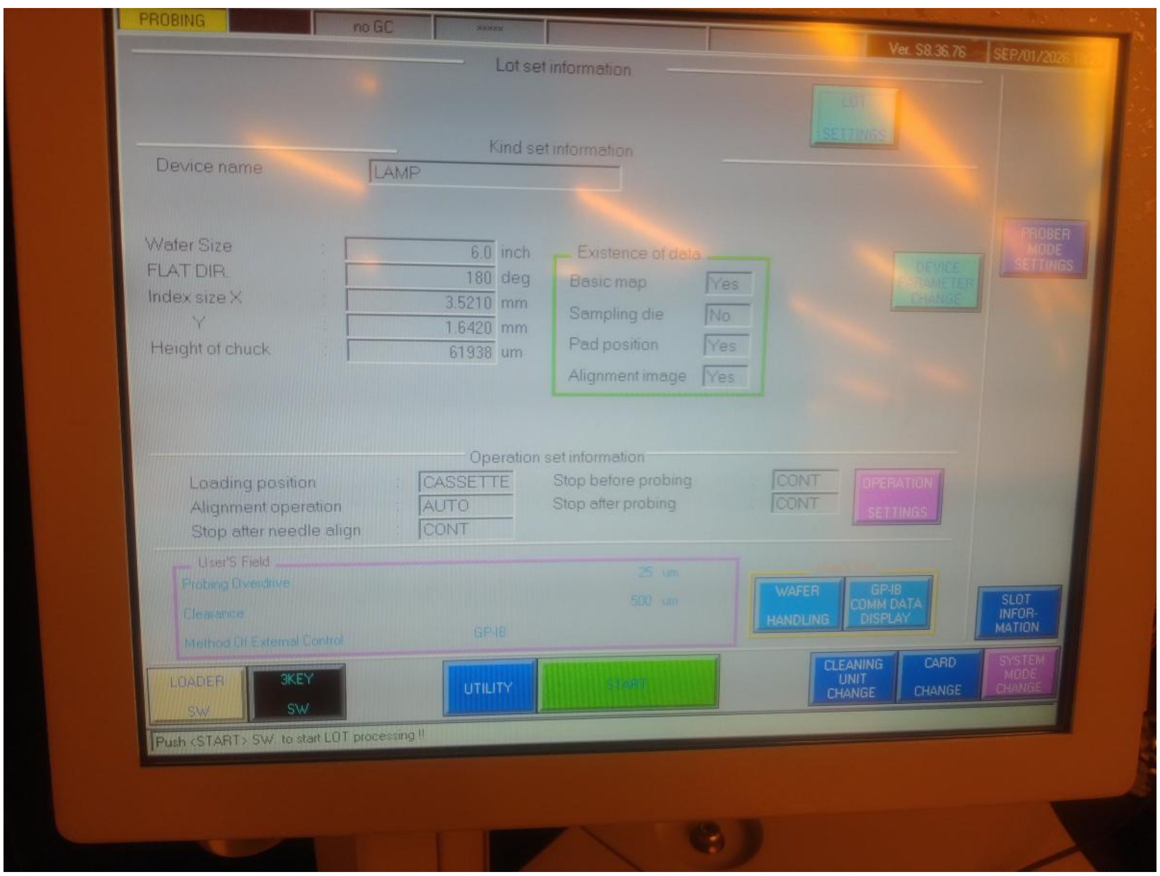
- Press "END" (**Figure 8o**) to complete the probe card change and return to the main production screen.

Figure 8o: END to Complete Card Change



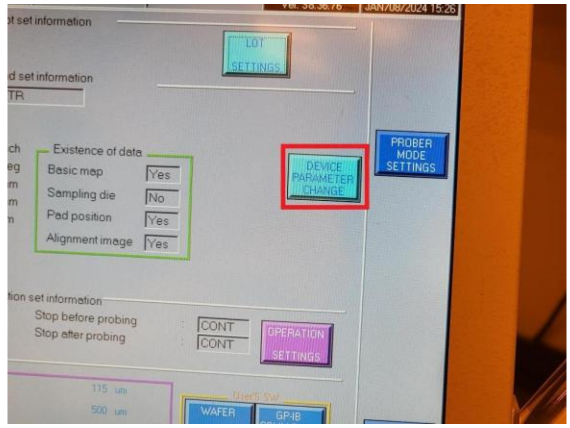
Changing Device Parameters

- Next, the correct recipe for the new probe card must be loaded. **If the device data is LAMP, you may skip this step.**



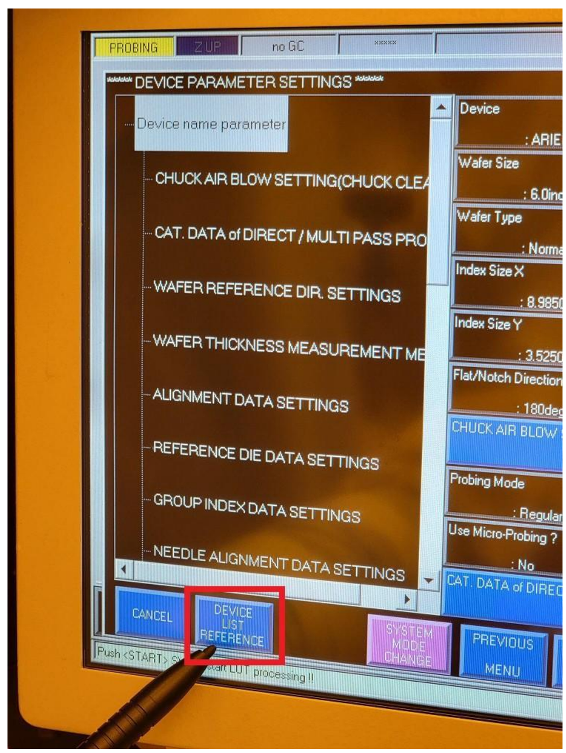
- Press “DEVICE PARAMETER CHANGE” (**Figure 8p**).

Figure 8p: DEVICE PARAMETER CHANGE



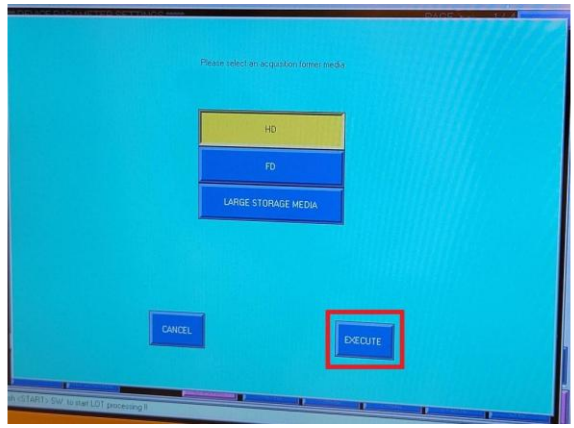
- Press “DEVICE LIST REFERENCE” (**Figure 8q**).

Figure 8q: DEVICE LIST REFERENCE



- Verify “HD” is highlighted and press “EXECUTE” (**Figure 8r**).

Figure 8r: EXECUTE

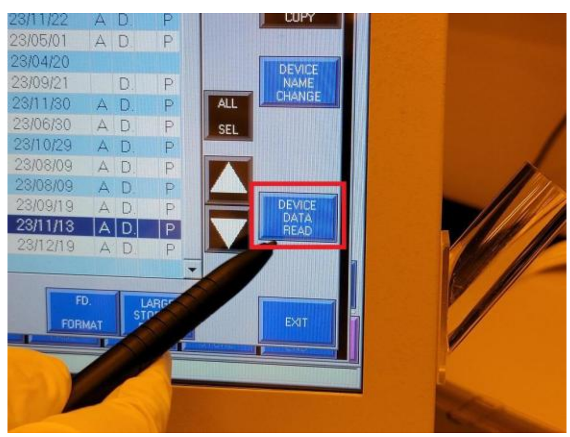


- Select the correct recipe for the new probe card. It is called LAMP on the list. (**Figure 8s**).

Figure 8s: Highlight the Intended Recipe in the Device List

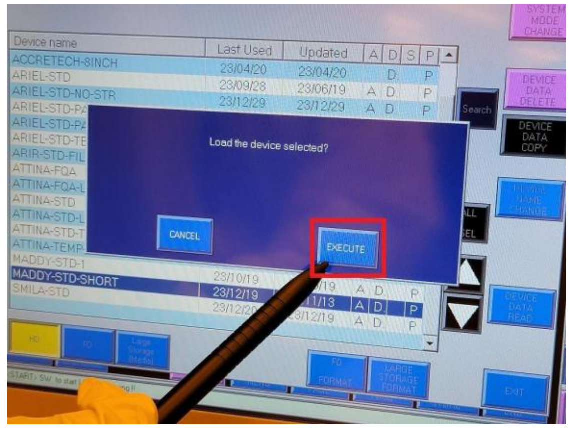
- Press “DEVICE DATA READ” (**Figure 8t**).

Figure 8t: DEVICE DATA READ



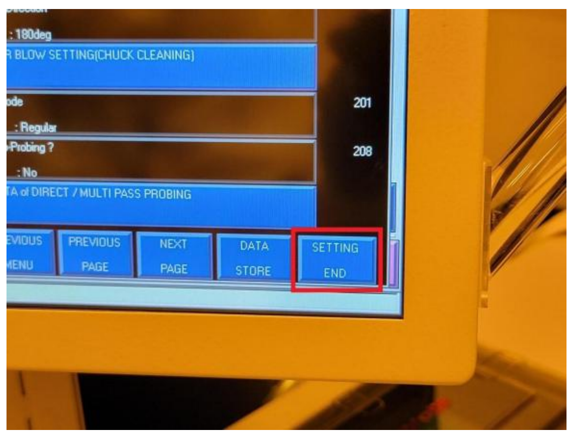
- When prompted, press “EXECUTE” (**Figure 8u**).

Figure 8u: EXECUTE



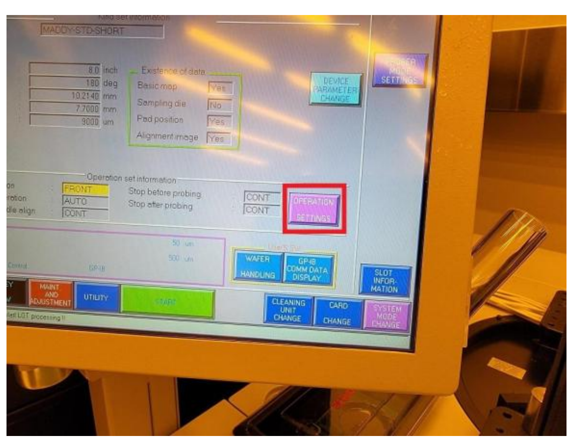
- Press “SETTING END” (**Figure 8v**) to return to the main production screen.

Figure 8v: SETTING END



- **Starting Probe Routine**
  - Press “OPERATION SETTINGS” (**Figure 8w**).

Figure 8w: OPERATION SETTINGS

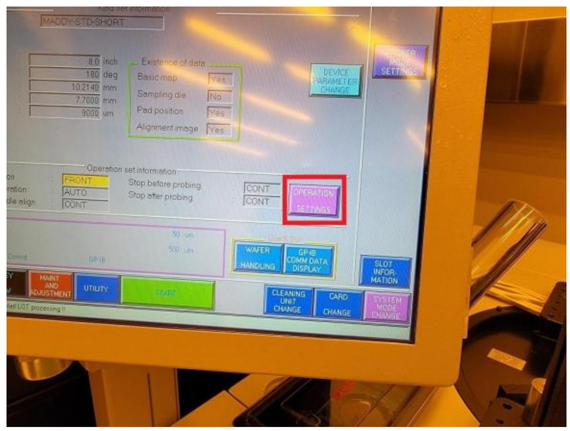


- Verify the correct Loading/Unloading position is highlighted and press “ENT” (**Figure 8x**).

  NOTE: The Loading/Unloading position will be cassette for lamp since it is 6’’, but can be loaded through front access door. For more details about front access check full manual.

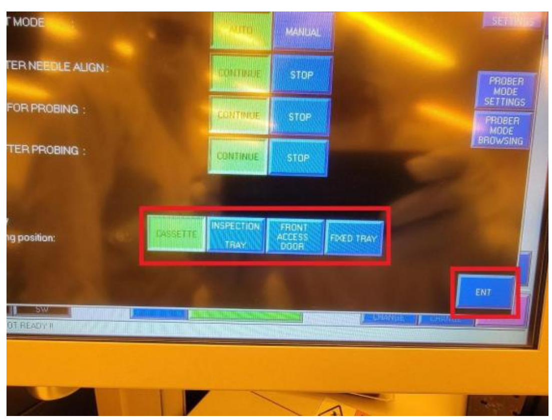
- **Cassette Loading**
  - Press “OPERATION SETTINGS” (**Figure 8at**).

Figure 8at: OPERATION SETTINGS



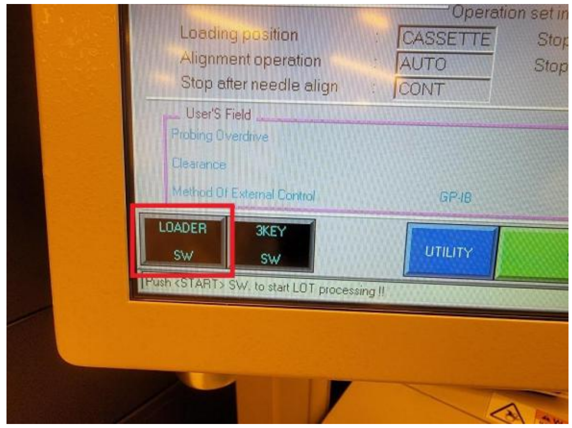
- Verify the “CASSETTE” Loading/Unloading position is highlighted and press “ENT” (**Figure 8au**).

Figure 8au: Verify CASSETTE Loading/Unloading Position



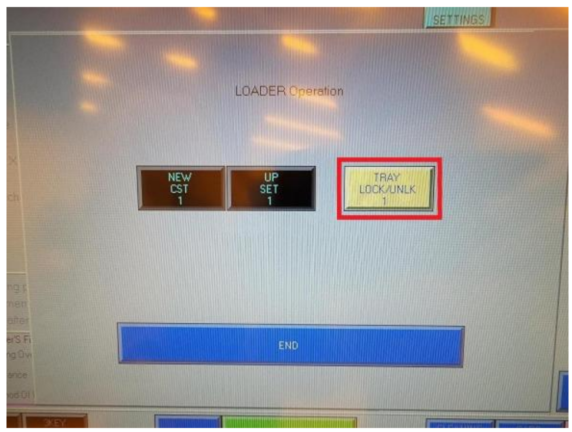
- Press “LOADER SW” (**Figure 8av**).

Figure 8av: LOADER SW



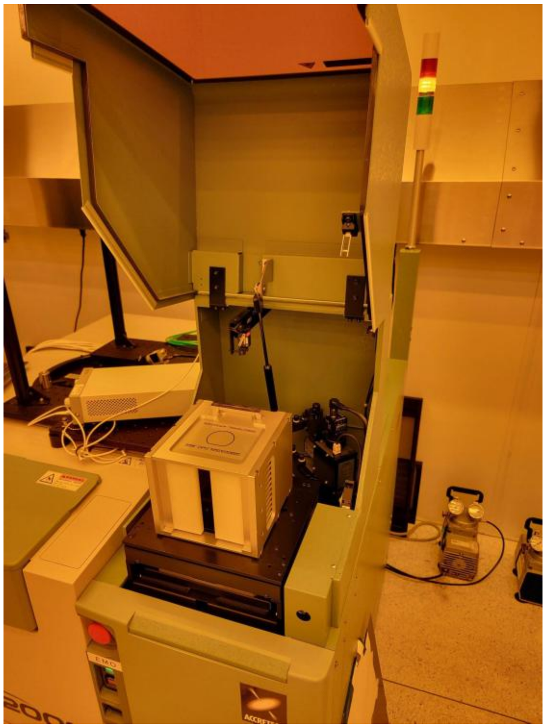
- Press “TRAY LOCK/UNLK 1” (**Figure 8aw**).

Figure 8aw: TRAY LOCK/UNLK 1



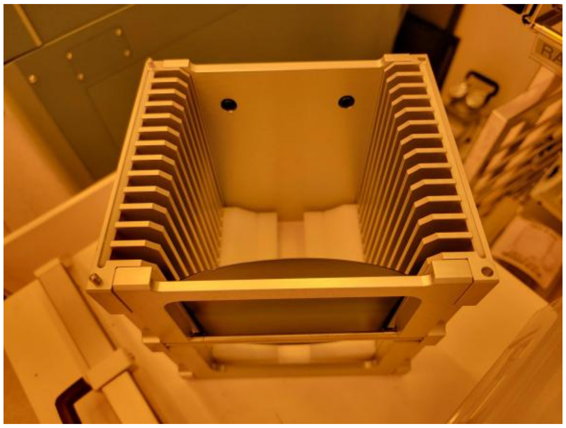
- Open the cassette tray (**Figure 8ax**).

Figure 8ax: Open Cassette Tray

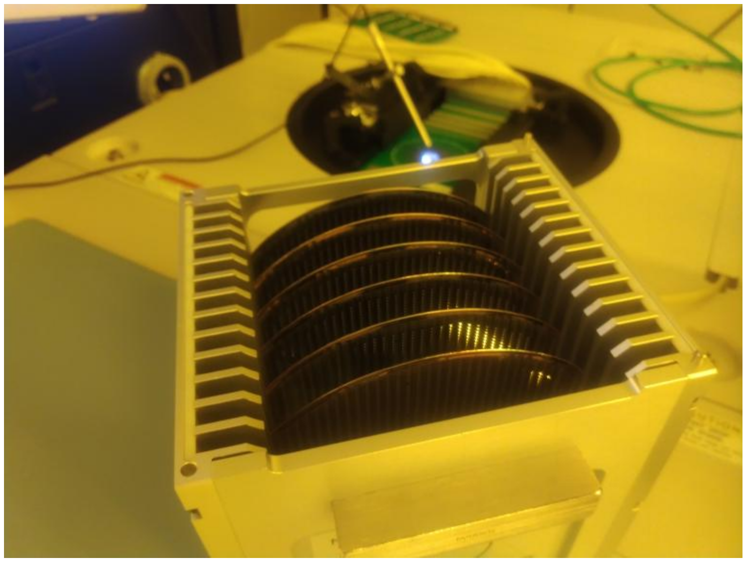


- Remove the **PROBE08** cassette and load wafers into consecutive slots starting from the slot closest to the H-bar. Load wafers so that the sides to be probed are facing away from the H-bar (**Figure 8ay**).

Figure 8ay: Wafer Loaded into Slot Closest to H-bar

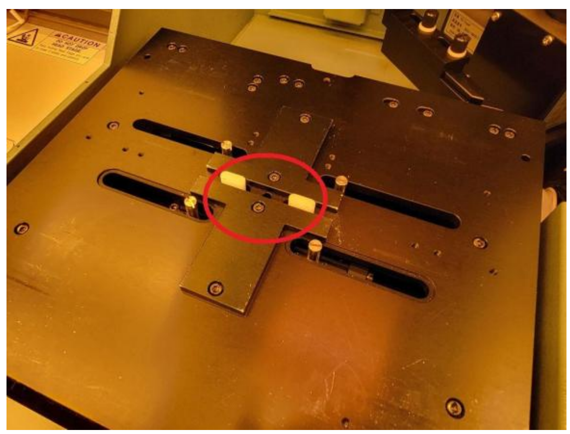


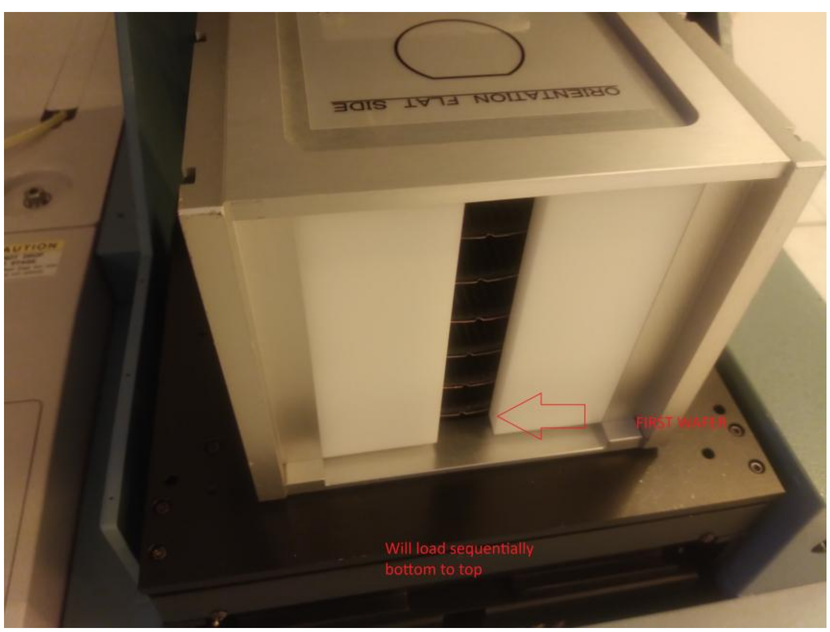
- If loading multiple wafers, it is recommended to space them apart to minimized wafer to wafer contact. Take extra care not to accidentally load them diagonally.



- Load the **PROBE08** cassette into the cassette tray so that the H-bar is facing down and fits within the groove (**Figure 8az**). Check that the notches are facing towards you and line them up to best of your abilities.

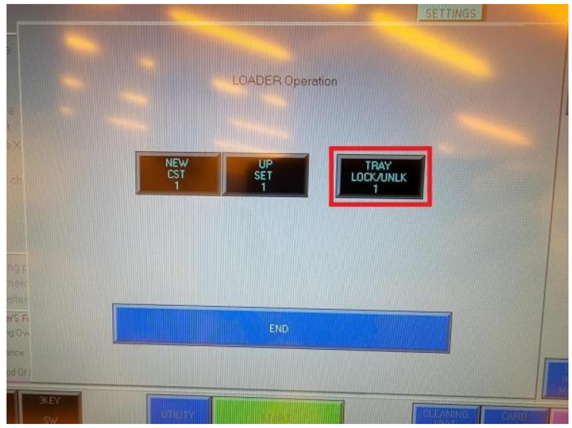
Figure 8az: Cassette Loaded into Tray





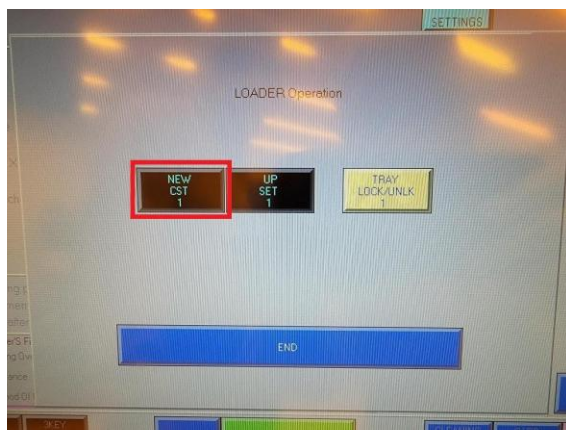
- Close the cassette tray.
- Press “TRAY LOCK/UNLK 1” (**Figure 8ba**).

Figure 8ba: TRAY LOCK/UNLK 1



- Press “NEW CST 1” (**Figure 8bb**).

Figure 8bb: NEW CST 1

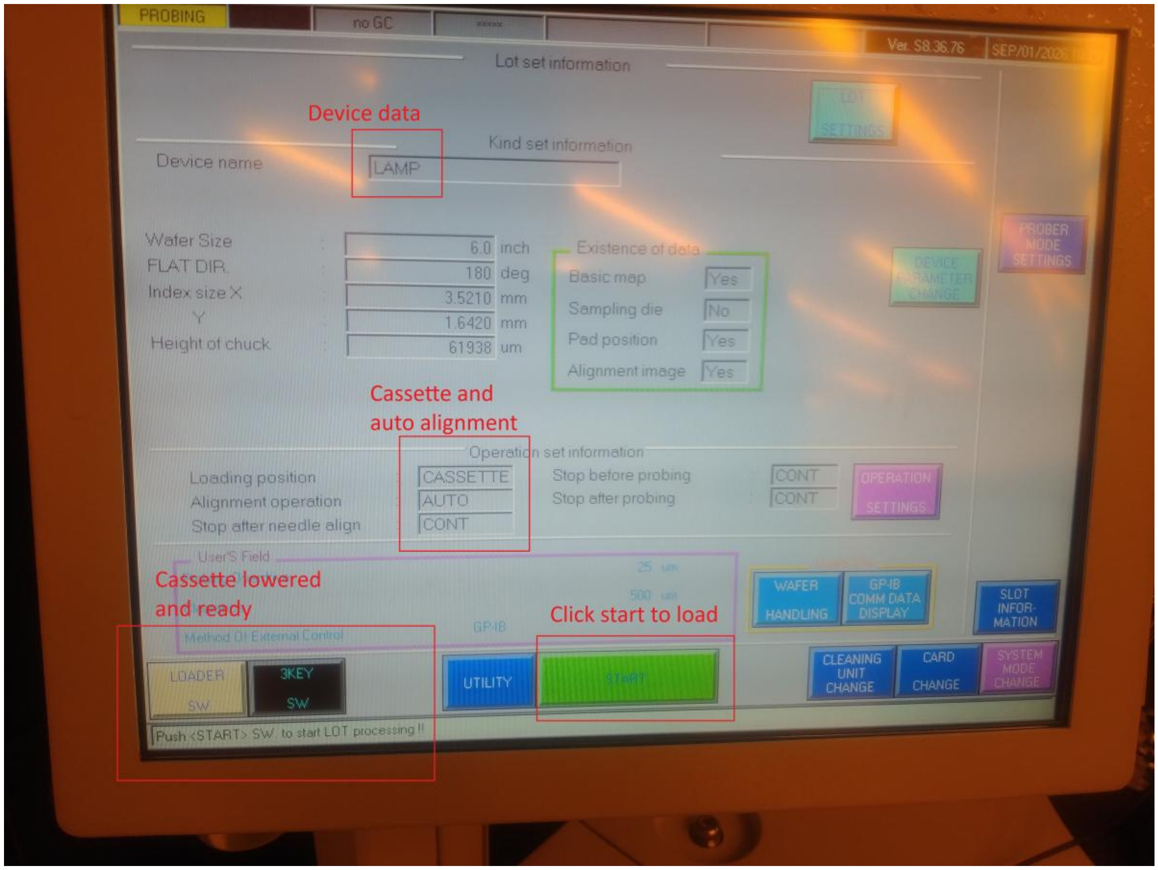


- The cassette will lower into the tool and a wafer will be loaded.

NOTE: If you need to raise the cassette, press “Up Set”. You cannot do this if loading operation is already started however.

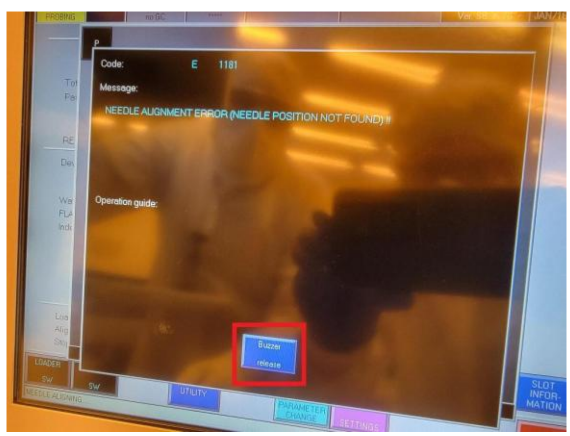
Figure 8ac: START

- Press Start once cassette is loaded, and the bottom left says press start to start lot processing. (CHECK that all fields are correct as shown below)



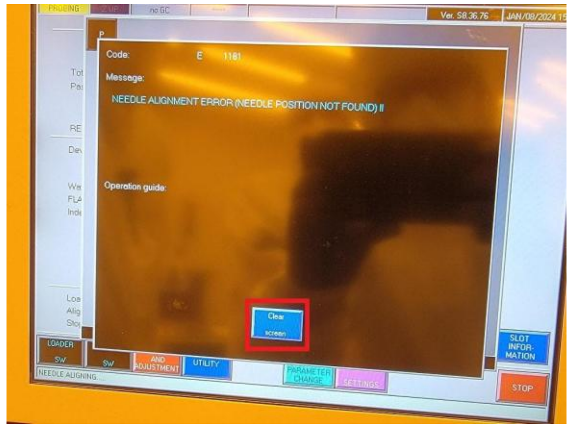
- For the first wafer, you may encounter needle alignment error. Press “Buzzer Release” when prompted with “NEEDLE ALIGNMENT ERROR” (**Figure 8ad**).

Figure 8ad: Buzzer Release



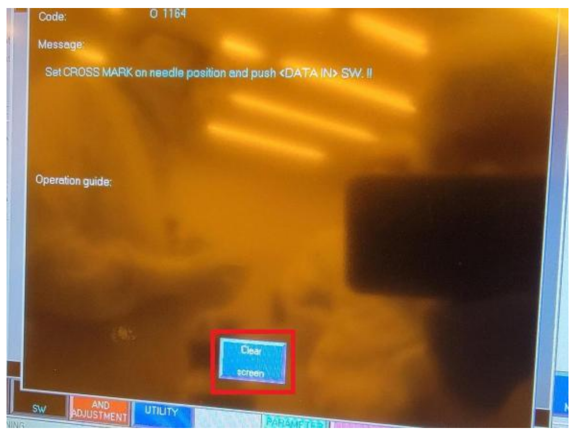
- Press “Clear screen” when prompted with another “NEEDLE ALIGNMENT ERROR” (**Figure 8ae**).

Figure 8ae: Clear Screen



- Press “Clear screen” when prompted with “Set CROSS MARK on needle position and push <DATA IN>” (**Figure 8af**).

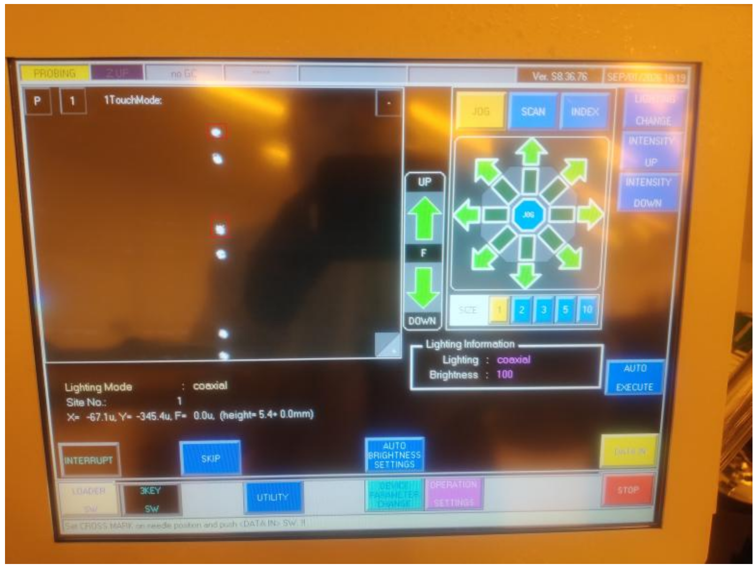
Figure 8af: Clear Screen



- Press the directional arrows to jog the chuck so that the tips of the probe needles are aligned with the circles overlaying the camera feed (**Figure 8ag**). **IMPORTANT: Make sure the two circles are aligned to the TOP LEFT pins. The top circle goes to the topmost pin, the bottom circle goes 3 pins down.**

NOTE: The dark green buttons can be used for fine adjustment.

Figure 8ag: Jog Chuck Until Needles Align with Circles

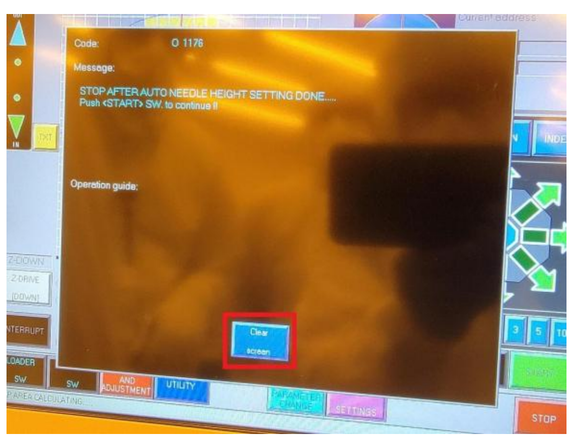


- Press “DATA IN” (**Figure 8ah**).

Figure 8ah: DATA IN

- Press “Clear screen” when prompted with “STOP AFTER AUTO NEEDLE HEIGHT SETTING DONE” (**Figure 8ai**).

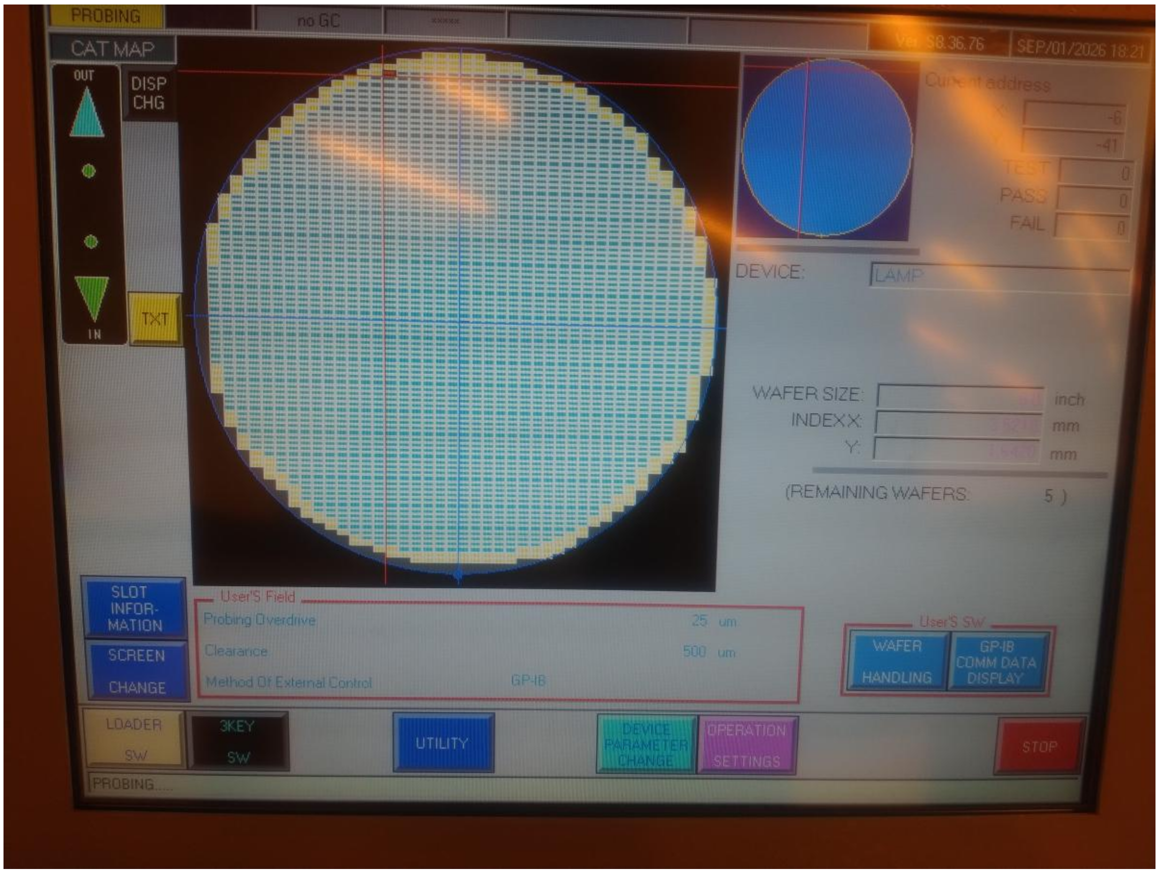
Figure 8ai: Clear Screen



- Verify the CARD RELATIVE Z HEIGHT and Probing Overdrive (**Figure 8aj**).

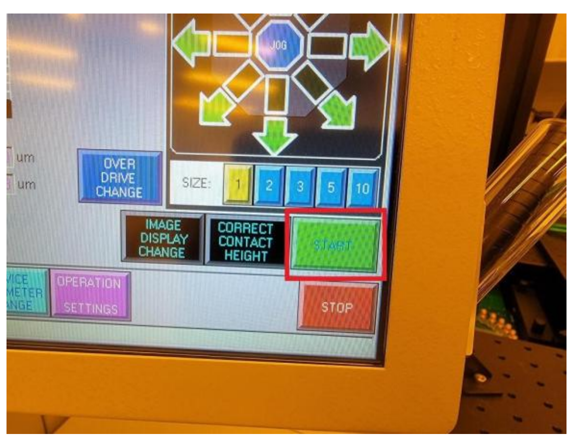
  NOTE: These are program specific settings. Contact Test/Metrology Engineer for correct settings.

Figure 8aj: Verify Probe Settings



- Press "START" (**Figure 8ak**).

Figure 8ak: START



- “Probing” will be displayed in the bottom left corner of the screen (**Figure 8al**).

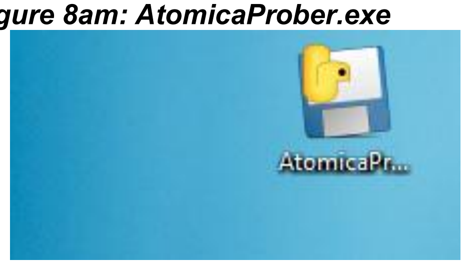
Figure 8al: Ready for Probing

Cassette Automation

- Move to the **PROBE08** desktop PC and launch the Atomica Prober executable. (**Figure 8am**).

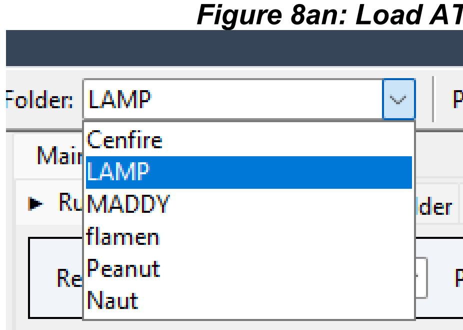
  NOTE: Login to the PC with the following credentials (User: Atomica, Password: Engineer2022)

Figure 8am: AtomicaProber.exe



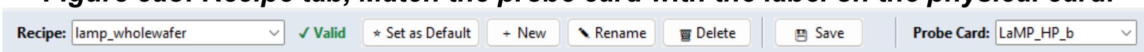
- Verify instruments are connected, verify working directory is correct. Load the LAMP ATA folder if not already loaded.

Figure 8an: Load ATA folder

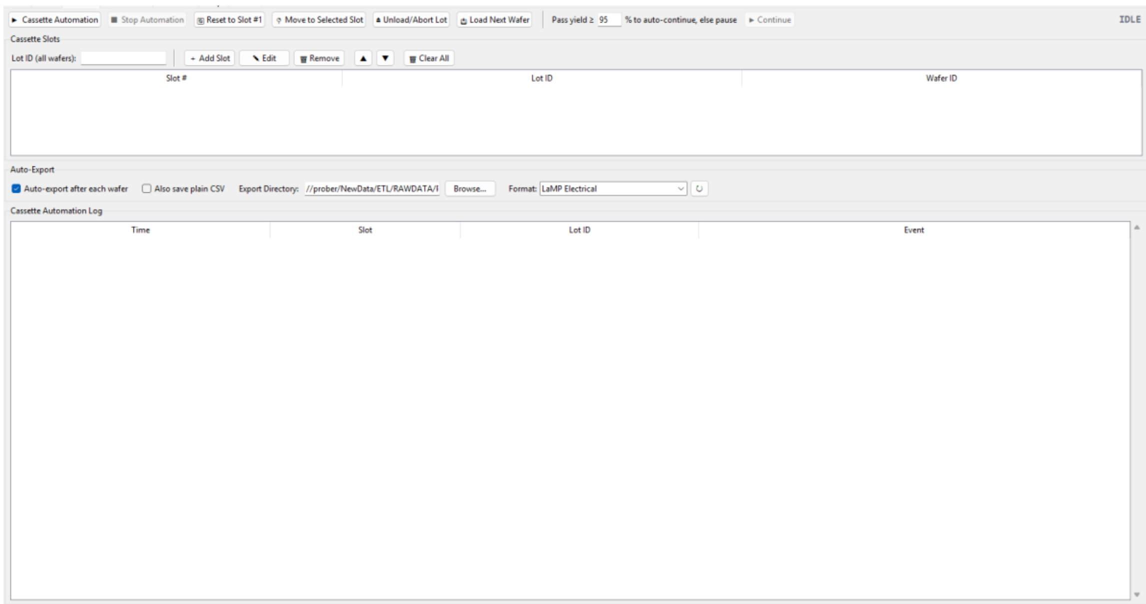


- Go to recipe tab, check that correct probe card is selected. Check that the recipe is correct (wholewafer for full wafer, electrical gauge if doing the gauge monitor). (**Figure 8ao**).

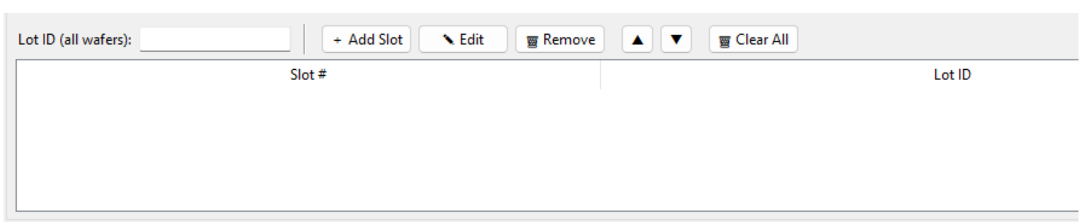
Figure 8ao: Recipe tab, Match the probe card with the label on the physical card.



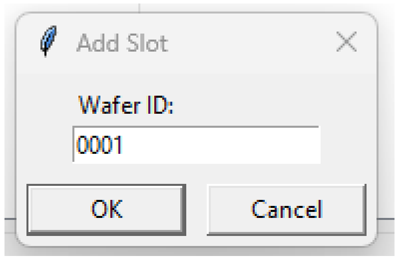
- **IF YOU HAVE MORE THAN ONE WAFER IN CASSETTE, Go to cassette tab to start automation. Otherwise skip.**



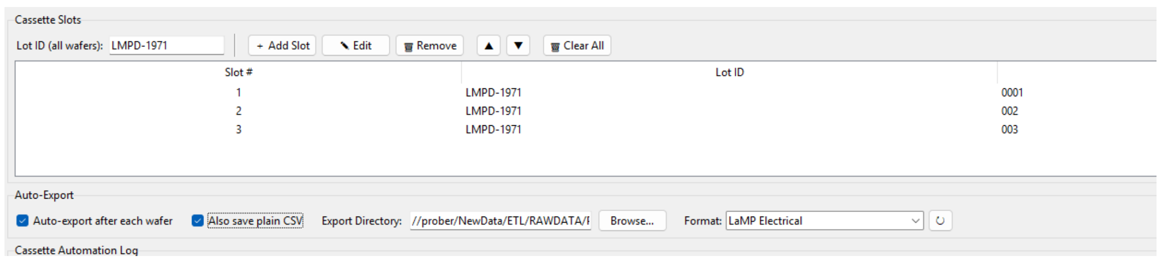
- This is cassette tab, here you can define the lamp and each wafer in the slots such that the whole lot can be automated and the data can be exported automatically.



- Fill in the lot id (e.g. LMPD-1921). Then for the wafer ids, click add slot, a window will popup.



- Fill them in one at a time, making sure that first slot is the bottom row of the cassette, because the prober loads wafers sequentially from bottom to top. (meaning the wafer closest to H bar is wafer 1).

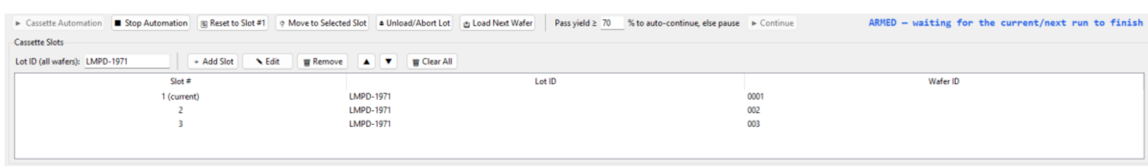


- Once you have it filled out, make sure auto export is checked. It is also recommended to have save plain csv, which makes it so you can import and view data afterwards.

- Check that the export directory is where you want it, (usually [\\prober\NewData\ETL\RAWDATA\PROBE08\Lamp](\\prober\NewData\ETL\RAWDATA\PROBE08\Lamp))

  Note: On the top right bar, you will see a pass yield percentage. It is there so that if a wafer has too many failing dies, the automation will pause and you will need to press the continue button after confirming everything looks ok, or if you want to skip, etc. You can change or reduce the percentage to 0, if you don't want this feature enabled.

- Once done, press Cassette Automation button. You should see it say "ARMED- waiting…" Now you still need to start the run for the first wafer, but it will take over afterwards and run the rest of the wafers in the same way as the first.

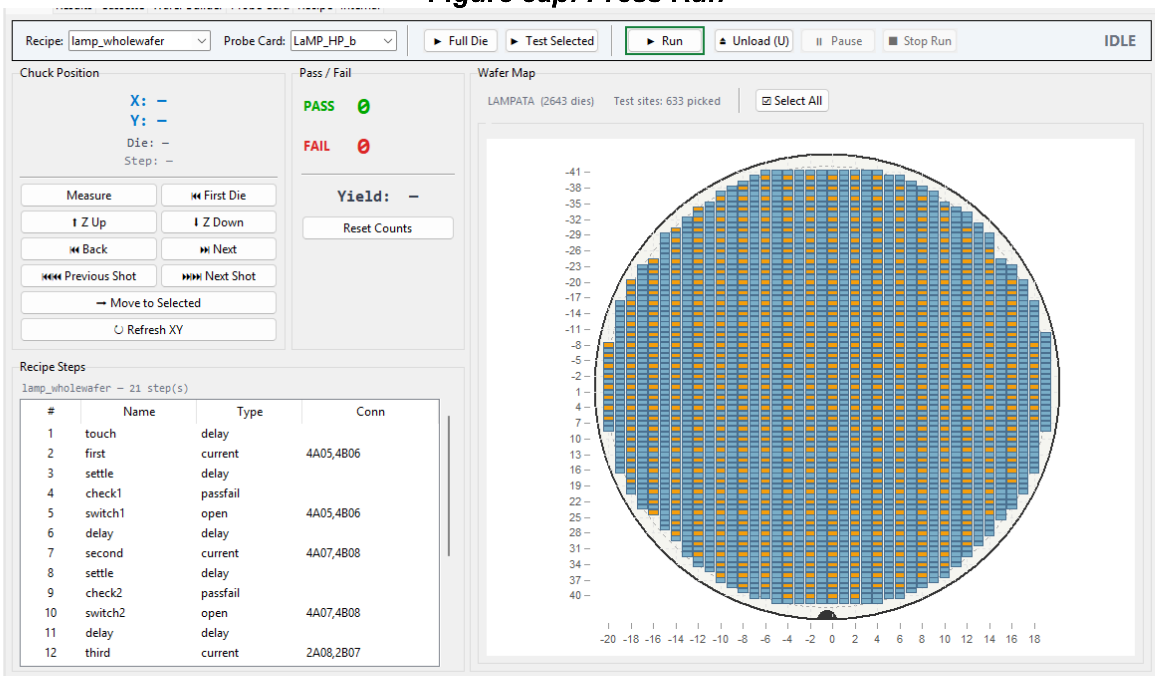


  Note: If you need to reset, click reset to slot 1. If you happen to skip a wafer manually on the prober (because of alignment error and you pressed load next wafer), then you can move to selected slot so that the code knows which slot it is on also. The prober and code is not guaranteed to be in sync about what slot it is running, check the prober to see how many wafers left to run. If you need to manually load next wafer or just unload the entire lot to abort, those buttons are included as well.

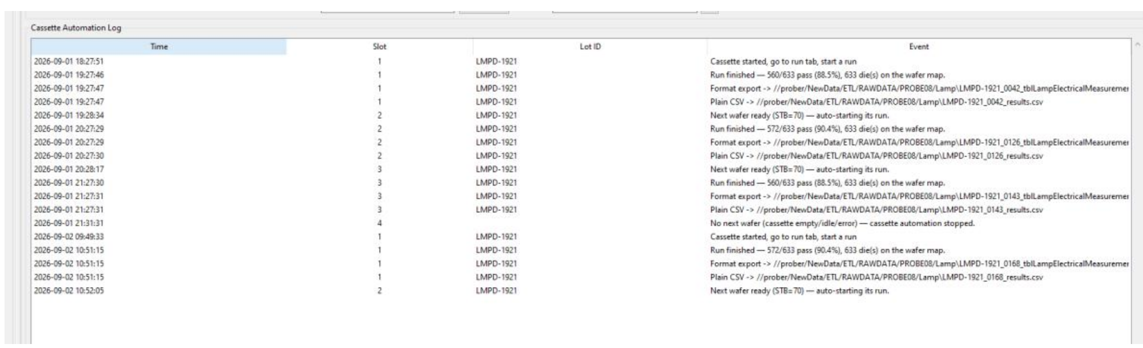
- Go to the Run tab, make sure recipe is loaded there as well. Press **RUN** to start.

Note: The prober will not run and throw an error unless it says “Probing” on the bottom left. (**Figure 8ap**).

Figure 8ap: Press Run



- Probing commences.
- If using cassette, you can check the log in the cassette tab, to see the current progress of the lot, or if there are any errors, to see why and whether you need to press continue. Example picture is provided below, also useful to check whether data has been exported by checking the output folder.



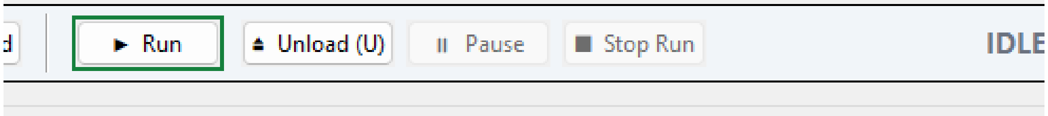
Cassette Automation Log

Time	Slot	Lot ID	Event
2026-09-01 18:27:51	1	LMPD-1921	Cassette started, go to run tab, start a run
2026-09-01 19:27:46	1	LMPD-1921	Run finished — 560/633 pass (88.5%), 633 die(s) on the wafer map.
2026-09-01 19:27:47	1	LMPD-1921	Format export -> //prober/NewData/ETL/RAWDATA/PROBE08/Lamp\LMPD-1921_0042_tblLampElectricalMeasuremer
2026-09-01 19:27:47	1	LMPD-1921	Plain CSV -> //prober/NewData/ETL/RAWDATA/PROBE08/Lamp\LMPD-1921_0042_results.csv
2026-09-01 19:28:34	2	LMPD-1921	Next wafer ready (STB=70) — auto-starting its run.
2026-09-01 20:27:29	2	LMPD-1921	Run finished — 572/633 pass (90.4%), 633 die(s) on the wafer map.
2026-09-01 20:27:29	2	LMPD-1921	Format export -> //prober/NewData/ETL/RAWDATA/PROBE08/Lamp\LMPD-1921_0126_tblLampElectricalMeasuremer
2026-09-01 20:27:30	2	LMPD-1921	Plain CSV -> //prober/NewData/ETL/RAWDATA/PROBE08/Lamp\LMPD-1921_0126_results.csv
2026-09-01 20:28:17	3	LMPD-1921	Next wafer ready (STB=70) — auto-starting its run.
2026-09-01 21:27:30	3	LMPD-1921	Run finished — 560/633 pass (88.5%), 633 die(s) on the wafer map.
2026-09-01 21:27:31	3	LMPD-1921	Format export -> //prober/NewData/ETL/RAWDATA/PROBE08/Lamp\LMPD-1921_0143_tblLampElectricalMeasuremer
2026-09-01 21:27:31	3	LMPD-1921	Plain CSV -> //prober/NewData/ETL/RAWDATA/PROBE08/Lamp\LMPD-1921_0143_results.csv
2026-09-01 21:31:31	4		No next wafer (cassette empty/idle/error) — cassette automation stopped.
2026-09-02 09:49:33	1	LMPD-1921	Cassette started, go to run tab, start a run
2026-09-02 10:51:15	1	LMPD-1921	Run finished — 572/633 pass (90.4%), 633 die(s) on the wafer map.
2026-09-02 10:51:15	1	LMPD-1921	Format export -> //prober/NewData/ETL/RAWDATA/PROBE08/Lamp\LMPD-1921_0168_tblLampElectricalMeasuremer
2026-09-02 10:51:15	1	LMPD-1921	Plain CSV -> //prober/NewData/ETL/RAWDATA/PROBE08/Lamp\LMPD-1921_0168_results.csv
2026-09-02 10:52:05	2	LMPD-1921	Next wafer ready (STB=70) — auto-starting its run.

Cassette Unloading

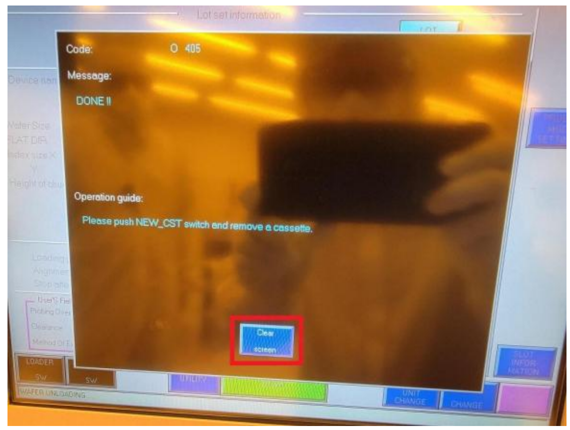
- Once probing has completed, press “UNLOAD” under the “Run” tab (or cassette tab) (**Figure 8aq**).
- If using more than one wafer in cassette, check in that tab that all wafers are done and exported.

Figure 8aq: UNLOAD



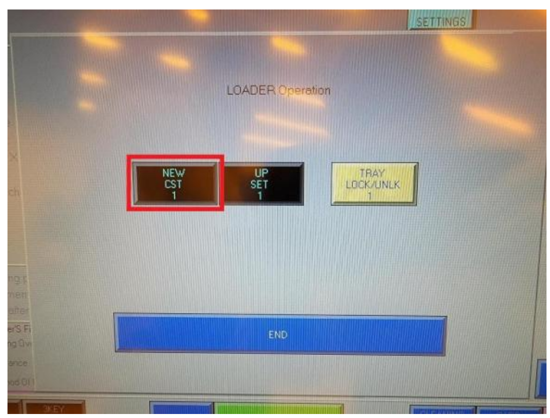
- After all wafers are unloaded, “Clear screen” when prompted with “DONE !!” (**Figure 8bc**).

Figure 8bc: Clear Screen



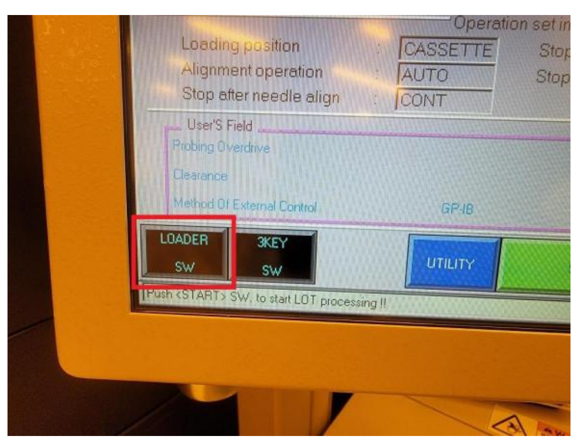
- Press “NEW CST 1” (**Figure 8bd**).

Figure 8bd: NEW CST 1



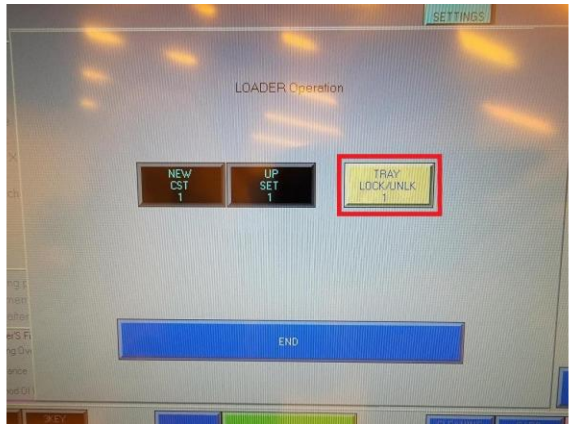
- Press “LOADER SW” (**Figure 8be**).

Figure 8be: LOADER SW



- Press “TRAY LOCK/UNLK 1” (**Figure 8bf**).

Figure 8bf: TRAY LOCK/UNLK 1

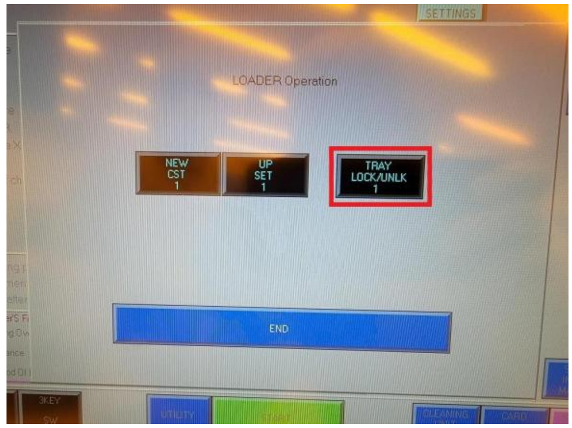


- Open the cassette tray and remove the wafers. Close the cassette tray with the empty **PROBE08** cassette loaded inside (**Figure 8bg**).

Figure 8bg: Leave Cassette Inside Tray

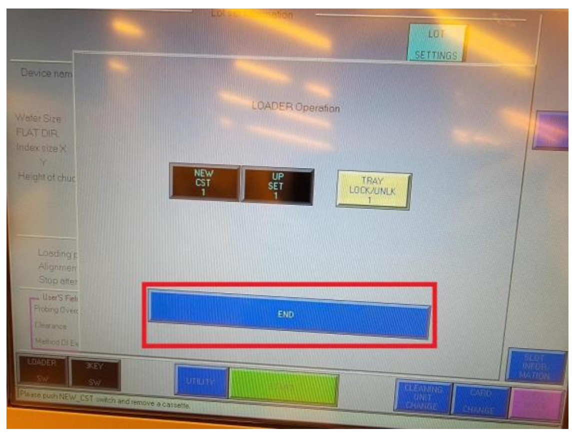
- Press “TRAY LOCK/UNLK 1” (**Figure 8bh**).

Figure 8bh: TRAY LOCK/UNLK 1



- Press “END” (**Figure 8bi**).

Figure 8bi: END



Qualification Procedure

Same procedure, use cassette loading, but no need for cassette automation. **IMPORTANT:** the device parameter on the prober is not LAMP, its LAMPGAUGE, the wafer will not load on the LAMP device data. Also, on the executable, make sure to run the Lamp Electrical Gauge Recipe, not whole wafer. Go to the \\prober\NewData\ETL\RAWDATA\PROBE08\Lamp to find the output data and load into SPC.